

Matplotlib Lab Program — Q31 & Q32

Question 31: Stacked Bar Plot with Error Bars

Aim

To write a Python program to create a stacked bar plot with error bars using Matplotlib.

Source Code

```
import matplotlib.pyplot as plt
import numpy as np

# Data
labels = ['G1', 'G2', 'G3', 'G4', 'G5']
men_means = [20, 35, 30, 35, 27]
women_means = [25, 32, 34, 20, 25]
men_std = [2, 3, 4, 1, 2]
women_std = [3, 5, 2, 3, 3]

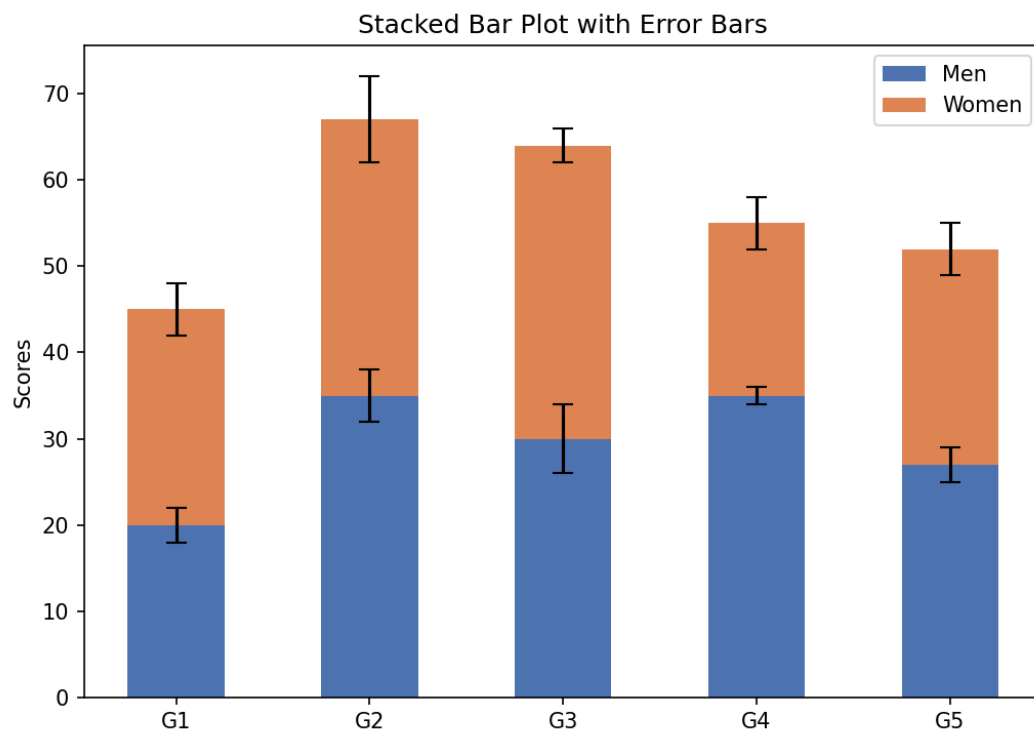
x = np.arange(len(labels))
width = 0.5

fig, ax = plt.subplots(figsize=(7,5))
ax.bar(x, men_means, width, yerr=men_std, label='Men', color='#4C72B0', capsize=5)
ax.bar(x, women_means, width, bottom=men_means, yerr=women_std, label='Women',
color='#DD8452', capsize=5)

ax.set_ylabel('Scores')
ax.set_title('Stacked Bar Plot with Error Bars')
ax.set_xticks(x)
ax.set_xticklabels(labels)
ax.legend()

plt.tight_layout()
plt.savefig('imgs/q31_output.png', dpi=150)
print("Stacked bar plot with error bars created and saved successfully.")
```

Output



Result

The Python program was executed successfully and a stacked bar plot with error bars was created, displaying the combined scores of two groups (Men and Women) across five categories along with their respective standard deviations.

Question 32: Scatter Graph of Random Distribution in X and Y

Aim

To write a Python program to draw a scatter graph by taking a random distribution in X and Y and plotting them against each other.

Source Code

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(42)
x = np.random.randn(100)
y = np.random.randn(100)

plt.figure(figsize=(7,5))
plt.scatter(x, y, color='#4C72B0', alpha=0.7)
plt.title('Scatter Plot of Random X and Y Distribution')
plt.xlabel('X - random values')
plt.ylabel('Y - random values')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig('imgs/q32_output.png', dpi=150)
print("Scatter plot of random distribution created and saved successfully.")
```

Output



Result

The Python program was executed successfully and a scatter graph was generated using 100 random points drawn from a standard normal distribution for the X and Y axes.